

New York Citys Traffic Collision Occurrences

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INTRODUCTION

In New York City, Americans rely solely on motor vehicles. Many folks need a motor vehicle to get by. Whether this motor vehicle is a sedan, a motorcycle, taxi, truck, or van, the need for a motor vehicle is highly recommended for everyday use. Transportation is a motor vehicle that gets from one location to another, such as going from points A to B en route.

A motor vehicle collision occurs every hour, day, month, and year in New York City. Roadways, highways, boroughs, on-streets, off-streets, cross-streets, zip codes, latitudes, and longitudes are associated with motor vehicle collisions. Many of these crashes result in a fatality, thereby resulting in awe.

Pictured right, a cab driver struck a cyclist and several pedestrians before hitting a building in the Flatiron district—photo by Jefferson Siegel for The New York Times.



OBJECTIVE

Aims to achieve good results, outcomes, and positive trend lines.

METHODOLOGY

A NYC OpenData Motor Vehicle Collisions - Crashes dataset was utilized. Preparing, reading, cleaning, pre-processing, and exploring the dataset were done to analyze the data where each row represented a crash event in a time series dataset. Exploratory data analysis was employed to make better decision-making, and a time series analysis was performed using time plots that were plotted against time observation.

ANALYSIS

Heatmap and Basemap were the maps designed to show where traffic collisions occurred in New York City. In a heatmap, the color denotes the accidents, creating specific little maps of the occurrences. In contrast, a base map was employed to diverge further into the map and see where the traffic collision was circled, squared, and pyramided, ranging from low to severe vehicle collisions. Moreover, pie charts, bar plots, and trend lines were shown to give a strong perspective of collisions on off-streets and boroughs. A time series analysis was performed, which displayed a beautiful line emerging from high crashes to low.

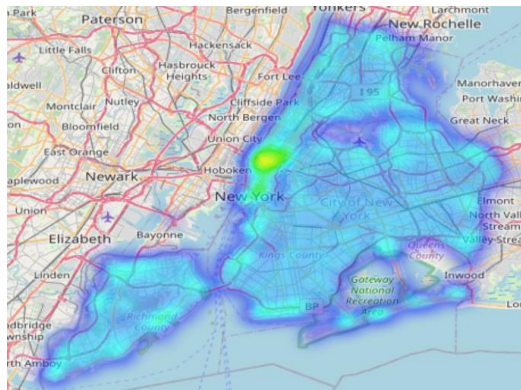


Fig 1: Heatmap of the Traffic Collisions

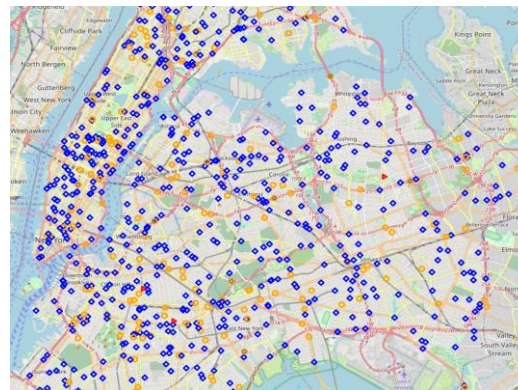


Fig 2: Basemap of the Traffic Collisions

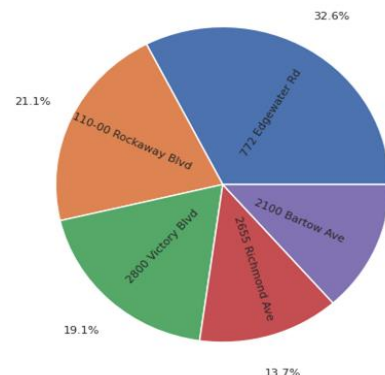


Fig 3: Top 5 Off-Street Names

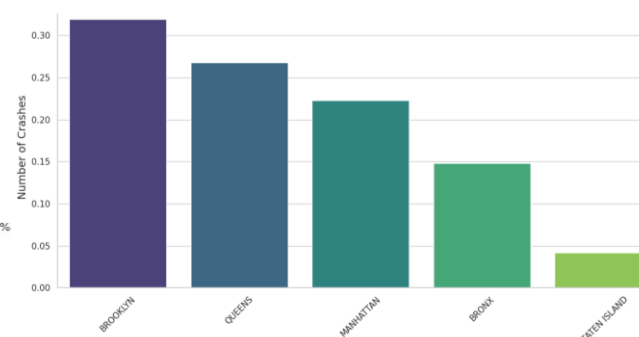


Fig 4: Top 5 Borough Names

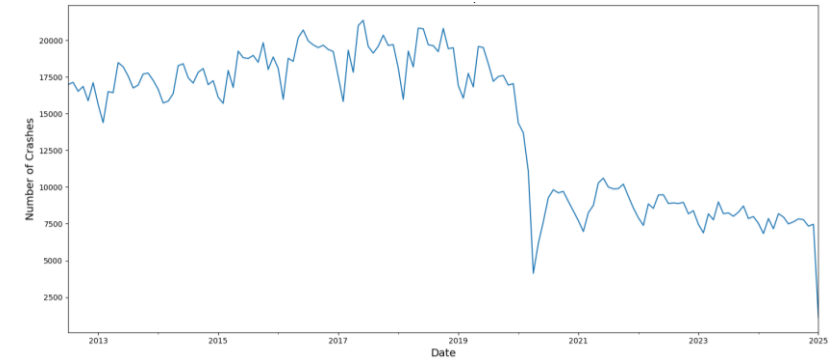


Fig 5: Time Series Analysis by the Number of Crashes per Month

CONCLUSION

Traffic collisions are on the rise in New York City, and to this day, traffic collisions continue to ensue daily, monthly, and yearly. In the heat map, the neon highlighted shows that many collisions occur throughout the boroughs, with the highest borough being Brooklyn in New York City, followed by Queens, Manhattan, Bronx, and Staten Island. In contrast to the heatmap, the base map portrays the traffic collisions by placing a mathematical shape, thereby making the map much more strategic. Off-street names were portrayed to show which off-street collisions occur, and there are many. A time series analysis of the number of crashes per month indicated a decline all because of COVID-19 when the people did not rely on vehicles and stayed home. Because of that, a declining trend was depicted and changes occurred, resulting in unpredicted fluctuation.

FUTURE WORK

Collect and analyze more data using Neural Networks and Machine Learning algorithms.

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REFERENCES

NYC Open Data, Northeast Big Data Innovation Hub, U.S. Department of Transportation Federal Highway Administration, and The New York Times.